

**Draft publication design · exact-file review required**

Complete source text follows. The source's own status and limitations remain in force; this edition does not establish execution, filing or approval.

Enterprise services, infrastructure and external dependencies

Version: 0.1.0 · **Planning horizon:** 2027–2031 · **PR:** #112

State: Implemented planning model. Owner-approved sourcing direction is pending canonical acceptance; numerical requirements and prices are proposals, not approved operating facts.

This package turns the September 9 decisions into a populated service inventory, shared-resource and supplier-dependency records, and an executable cost/capacity/staffing comparison. It does not deploy infrastructure, purchase anything, alter payroll, or rewrite existing finance releases.

Registers and inputs

File	Contents
Service register	57 service records, actual work, accountable roles, business recipients, sourcing, linked support, dependencies and CCF controls.
Internal support register	49 teams/functions, facilities and environments. Design requirements are separated from existing functions and unknown occupied capacity.
External counterparty/dependency register	Four source-named parties and 49 service-dependency requirements. Unselected suppliers remain null; internal ARU-to-Red Wash service is separately identified.
Workload and recovery requirements	All six hosting classes; proposed demand, recovery targets, dependency order and failure-domain tests.
Economics and transition inputs	Explicit synthetic rates, workloads, seat mix, staff-effort assumptions, asset lives, alternatives, sensitivities, exclusions and staged ownership gates.
Source inventory and finance extract	Reviewed main/PR revisions, source paths/blob checks, states and the conditional finance-policy extract.

The service and dependency files use columns plus row arrays to keep diffs manageable. Their common defaults are explicit in the same files. The loader expands every field and generates ordinary JSON and CSV registers. null means unknown/not approved, not zero or not applicable. Source definitions, expanded exports and database tables have distinct roles; editing generated exports does not change inputs.

Run

Python 3.11 or newer; the service model itself uses only the standard library. Run from the repository root:

```
python -m enterprise.services.model validate --repository-root .
python -m unittest discover -s enterprise/services/tests -v
python -m enterprise.services.model build --repository-root . --output /tmp/sable-services-run --
database
```

Use a fresh output directory when requesting SQLite: the builder refuses to silently replace an existing database. Omit --repository-root only for isolated model tests; that mode does not claim to have checked repository source bytes or CCF/site references.

Outputs include expanded registers, monthly costs, staffing/allocations, SaaS cost lines, capacity, a finance bridge, comparison CSV/JSON, 45 demand/price sensitivity results, a readable report, a checksum manifest and optional SQLite with foreign-key-enforced service/component/dependency links. The manifest excludes SQLite bytes because database packaging is not a canonical identity; source and tabular outputs provide reproducibility.

Example local query:

```
SELECT s.name, s.sourcing, d.id, d.provider_id
FROM service AS s
JOIN service_dependency AS sd ON sd.service_id=s.id
JOIN dependency AS d ON d.id=sd.dependency_id
WHERE d.provider_id IS NULL
ORDER BY s.name, d.id;
```

Comparison scope

Four architecture alternatives use the same logical service demand: cloud-heavy transition; mixed cloud/owned colocation; owned baseline in professional colocation; and eventual owned primary physical plant with independent recovery still in colocation. Every alternative preserves internally controlled restricted infrastructure and local equipment. A fifth scenario models an illustrative transition at months 7 and 19. Those months are analytical assumptions, not promised installation dates; the recorded acceptance gates control execution.

Prices are analyst-entered synthetic comparison assumptions, **not researched tariffs, vendor quotes or commitments**. Headcount, demand, growth, license mix, workload effort and numerical recovery targets must be validated before procurement or staffing approval. No preferred architecture is declared cheapest in the real world from this illustrative model.

The generated comparison and report are reproducible examples, not new canon. The full outputs are generated by the CI workflow and attached to its run as build artifacts, not represented as a controlled release package.

Accounting and workforce safeguards

The existing 2027 conditional policy retains 48 occupied / 54 authorized ESS positions. Its \$216,000 annual vendor allowance, \$1,140,000 annual corporate facilities allowance and \$6,720,000 ESS payroll are displayed in the bridge. They are **not automatically deducted** from the new model. Actual reused staff, mapped overlap and generated product-compute expense require source-level reconciliation before a net successor forecast can be accepted.

The model reads contract input exposure separately and labels it before churn/renewal/engine adjustments. That exposure is not booked compute expense and is not used as a replacement credit. Original policy, assets, industrial books, released forecasts and intercompany eliminations are not rewritten.

Shared team costs are counted once and allocated across workloads. Required fractional capacity, rounded dedicated positions, reusable staff, contractors and approved hiring are distinct. A hypothetical reuse scenario is not evidence of available people. J2/Atlas personnel are not treated as a spare platform workforce. Education billets are not added to J2 a second time.

Capital cash, operating cash, depreciation, net book value, refresh and exit costs are separate. Depreciation starts the month after purchase; refresh is not simultaneously expensed as a reserve. Owned physical plant is capacity-bounded and has a separate useful life and facilities workforce.

Boundaries and next acceptance gates

Read operating assumptions and acceptance gates before interpreting targets or totals. Particularly important: the local hardware envelope is not a complete resilient OT design; N+1 CPU calculations do not prove storage/GPU redundancy, carrier independence or RTO/RPO; SaaS availability does not establish a customer-controlled recoverable runtime; and immutable online and offline backup controls require separate evidence.

Authority remains with the [sourcing decision record](#), its [scope supplement](#), and accepted Technology/ESS/J2/business doctrine. PR acceptance is distinct from implementing or funding an operating estate.

September 11 runtime successor

The optional versioned `runtime layer` is loaded from the two `runtime_*_2026-09-11.json` inputs. It updates selected counterparties and planned placements, emits `runtime.json`, and joins the separate enterprise finance and geography successors. The older alternative-cost report remains a preserved comparison, not the owned-facility investment decision.